

The Geometry of Real-Valued Functions

Functions, Graphs, and Level Sets (Curves & Surfaces)

Abhirup Moitra

Aim of these notes: We develop a geometric language for understanding functions by visualizing them. In particular, we formalize:

(i) Graphs (ii) Level Sets / Level Curves (iii) Level Surfaces.

These lecture notes are written as a guided introduction to the geometry of real-valued functions and the visual language that supports multivariable calculus. The main goal is to build a clean bridge between the formal set-theoretic definition of a function and the geometric objects that arise from it: graphs, level sets, contour maps, and sections. While the ideas are classical, the emphasis here is deliberately structural and rigorous: every picture is treated as a mathematical object that can be described (and verified) by definitions, equivalences, and proofs.

(Inspired by Marsden & Tromba: Vector Calculus.)

Contents

1	Functions and Mappings	3
1.1	Foundational Viewpoint	3
1.2	Relations, Functions, and The Uniqueness Axiom	3
1.3	Domain, Codomain, Range/Image	4
1.4	Scalar-Valued and Vector-Valued Functions	4
1.5	Rigorous Analysis of The Examples	5
1.6	A Short Lemma Connecting to Later Geometry	6
1.7	A Mapping Diagram	7
2	Graphs of Functions	7
2.1	Definition of the Graph	7
2.2	Basic Lemmas About Graphs	7
2.3	Characterization Theorem: When is a Set a Graph?	8
2.4	Rigorous Solved Example (Graph Test)	8
3	Level Sets, Curves, and Surfaces	9
3.1	Motivation	9
3.2	A Standard Geometric Picture	9
3.3	Contour Maps (Level Curves) for a Height Function	10
3.4	Example 1 (Rigorous): Constant Function and Its Level Curves	10
3.5	Example 2 (Rigorous): Linear Function and Its Level Curves	11
4	Level Curves and Surfaces	12
4.1	Definitions and Basic Principles	12
4.2	Example: The Quadratic Function $f(x, y) = x^2 + y^2$	14
4.2.1	Solution	14
5	The Method of Sections	15
5.1	Foundational Definitions	15
5.2	A Rigorous Lemma for Graph Sections	16
5.3	Example 4: The Hyperbolic Paraboloid (saddle)	17
5.4	Example 5: Level Surfaces of $x^2 + y^2 + z^2$	18
5.4.1	Solution	18
5.5	Example 6: Level Surfaces of $x^2 + y^2 - z^2$	19
5.5.1	Solution	19
6	Method of Sections for Graphs of Functions of Three Variables	19
6.1	Graphs in $\mathbb{R}^4$ and why we use sections	19
6.2	A formal lemma describing coordinate sections	20
6.3	Worked example in full detail: $f(x, y, z) = x^2 + y^2 - z^2$	21
6.4	Practice Problems	23
7	Conclusion	27

Preface

Our development is motivated by, and follows the spirit of, the early geometric discussion in *Vector Calculus* by Jerrold E. Marsden and Anthony Tromba. In particular, the viewpoint adopted throughout is that a function is not merely a formula but a mapping between spaces, and that the geometry of a function is best understood by studying the subsets it determines in Euclidean space. We begin with foundational notions: domains and ranges, scalar-valued versus vector-valued mappings, equality of vector-valued functions via component functions, and the graph as a subset of a product space. From there we develop the geometry of graphs in $\mathbb{R}^{n+1}$, with the “vertical line test” expressed as a precise set-theoretic criterion for when a set is the graph of a function.

A central theme of these notes is that *level sets* encode the shape of a graph. For functions of two variables, level curves (contours) provide a topographic description of surfaces; for functions of three variables, level surfaces generalize this idea to three-dimensional domains. We state and prove the fundamental relationship between horizontal slices of a graph and their projections onto the domain, which explains why contour maps faithfully represent the geometry of a surface. When graphs live in higher-dimensional spaces (for example, graphs of functions $\mathbb{R}^3 \rightarrow \mathbb{R}$ lie in $\mathbb{R}^4$), we rely on the method of sections: slicing by coordinate hyperplanes produces three-dimensional surfaces that can be visualized and analyzed rigorously.

Alongside the mathematics, we also adopt a computational viewpoint: visualizations are treated as faithful translations of definitions. To that end, we include Python-based models (Matplotlib and Plotly) that reproduce the geometric structures discussed in the text—wireframes, level curves lifted to surfaces, and multi-view projections that clarify how perspective affects interpretation. The intention is not to replace theory with computation, but to use computation as a verification tool and as an additional lens for geometric insight.

These notes are designed for readers who want both conceptual clarity and proof-level precision. Each section aims to move from **Definition** to **Lemma/Theorem** to **Proof**, and then to worked examples and practice problems. The exercises are chosen to reinforce a single message: geometry in multivariable calculus is not a collection of pictures, but a collection of rigorously defined sets whose relationships can be proved and visualized. Any errors remain the author’s responsibility, and the presentation is meant to be iteratively improved through practice, computation, and revisiting the foundational definitions.

Online Access (Book Link)

For convenient online access to the course reference book, see the following link:

[Vector Calculus \(Marsden & Tromba\) -- Google Drive Access](#)

Software and Practice Resources

If you would like to see exactly how the figures were generated (including the multi-projection layouts and the construction of slice curves), and to practice by modifying parameters and extending the examples, please consult the accompanying GitHub repository:

[Functions, Graphs, and Level Sets \(Curves & Surfaces\)-- Computational Guide](#)

1 Functions and Mappings

1.1 Foundational Viewpoint

Before defining graphs, it is conceptually helpful to recall that any subset $R \subseteq U \times \mathbb{R}$ may be viewed as a *relation* between inputs $x \in U$ and outputs $y \in \mathbb{R}$. A *function* is a special kind of relation.

1.2 Relations, Functions, and The Uniqueness Axiom

Definition: Cartesian Product and Ordered Pairs

For sets A and B , the *Cartesian product* is

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

An *ordered pair* (a, b) is characterized by the property:

$$(a, b) = (a', b') \iff (a = a' \text{ and } b = b').$$

Definition: Relation

Let U and V be sets. A *relation* from U to V is any subset

$$R \subseteq U \times V.$$

For $x \in U$, the *fiber* (or *image-set*) of R at x is

$$R(x) := \{y \in V : (x, y) \in R\}.$$

Definition: Function As a Special Relation

Let U and V be sets. A *function* (or *mapping*)

$$f : U \rightarrow V$$

is a relation $f \subseteq U \times V$ satisfying the two axioms:

(Existence) For every $x \in U$, there exists $y \in V$ such that $(x, y) \in f$.

(Uniqueness) For all $x \in U$ and all $y_1, y_2 \in V$,

$$(x, y_1) \in f \text{ and } (x, y_2) \in f \implies y_1 = y_2.$$

Equivalently,

$$\forall x \in U \exists! y \in V \text{ such that } (x, y) \in f,$$

and then we write $y = f(x)$.

Remark: Set-Theoretic Viewpoint

A function can be identified with its graph as a subset of $U \times V$, but we will keep the notation $f : U \rightarrow V$ and interpret $(x, f(x))$ as the corresponding pair in $U \times V$.

1.3 Domain, Codomain, Range/Image

Definition: Domain, codomain, and range (image)

Let $f : U \rightarrow V$ be a function.

- The set U is the *domain*, denoted $\text{dom}(f) := U$.
- The set V is the *codomain* (or *target space*).
- The *range* (or *image*) is

$$\text{ran}(f) := f(U) := \{f(x) : x \in U\} \subseteq V.$$

Remark: Range vs. codomain

Always remember:

$$\text{ran}(f) \subseteq \text{codomain}.$$

The codomain is chosen as part of the definition of the mapping, whereas the range is determined by the actual outputs.

Proposition: Characterization of The Image Set

Let $f : U \rightarrow V$ and let $S \subseteq U$. Define $f(S) := \{f(x) : x \in S\} \subseteq V$. Then for any $y \in V$,

$$y \in f(S) \iff \exists x \in S \text{ such that } f(x) = y.$$

Proof. ($\Rightarrow$) If $y \in f(S)$, then by definition of $f(S)$ there exists $x \in S$ with $y = f(x)$.

($\Leftarrow$) If there exists $x \in S$ with $f(x) = y$, then y belongs to the set $\{f(x) : x \in S\} = f(S)$. $\square$

1.4 Scalar-Valued and Vector-Valued Functions

Definition: Scalar-valued vs. vector-valued maps

Let $U \subseteq \mathbb{R}^n$ and let $m \in \mathbb{N}$. A map

$$f : U \rightarrow \mathbb{R}^m$$

is called:

- *scalar-valued* if $m = 1$ (so $f : U \rightarrow \mathbb{R}$),
- *vector-valued* if $m > 1$.

Definition: Component Functions

If $f : U \rightarrow \mathbb{R}^m$ is vector-valued, we write

$$f(x) = (f_1(x), f_2(x), \dots, f_m(x)),$$

where each $f_i : U \rightarrow \mathbb{R}$ is a scalar-valued function called the i -th *component* of f .

Proposition: A Vector-Valued Map is Determined by Its Components

Let $f, g : U \rightarrow \mathbb{R}^m$ with

$$f = (f_1, \dots, f_m), \quad g = (g_1, \dots, g_m).$$

Then

$$f = g \iff f_i = g_i \text{ for all } i = 1, \dots, m.$$

Proof. ($\Rightarrow$) If $f = g$, then for every $x \in U$ we have $f(x) = g(x)$ as vectors in $\mathbb{R}^m$. Equality of vectors means equality of each coordinate, hence $f_i(x) = g_i(x)$ for every i and every x . Thus $f_i = g_i$ as functions.

($\Leftarrow$) Assume $f_i = g_i$ for all i . Then for any $x \in U$,

$$f(x) = (f_1(x), \dots, f_m(x)) = (g_1(x), \dots, g_m(x)) = g(x).$$

Therefore $f = g$. □

1.5 Rigorous Analysis of The Examples

Example: Example 1: A Scalar-Valued Function on $\mathbb{R}^3 \setminus \{0\}$

Define $f : \mathbb{R}^3 \setminus \{(0, 0, 0)\} \rightarrow \mathbb{R}$ by

$$f(x, y, z) = (x^2 + y^2 + z^2)^{-3/2}.$$

Step 1: Domain: The expression involves a negative exponent, so we must exclude points where

$$x^2 + y^2 + z^2 = 0.$$

But in $\mathbb{R}^3$ we have $x^2 + y^2 + z^2 = 0$ if and only if $x = y = z = 0$. Hence the correct domain is

$$\text{dom}(f) = \mathbb{R}^3 \setminus \{(0, 0, 0)\}.$$

Step 2: Well-Definedness (Existence and Uniqueness of Output): Fix $(x, y, z) \in \text{dom}(f)$. Then $x^2 + y^2 + z^2 > 0$, so

$$(x^2 + y^2 + z^2)^{-3/2}$$

is a uniquely determined real number (since the real power $t^{-3/2}$ is defined for every $t > 0$). Therefore $f(x, y, z)$ exists and is unique for every input in $\text{dom}(f)$, so f is a well-defined function.

Step 3: Range Information: For every $(x, y, z) \in \text{dom}(f)$ we have $x^2 + y^2 + z^2 > 0$, so $f(x, y, z) > 0$. Moreover, letting $r = \sqrt{x^2 + y^2 + z^2}$, we can rewrite

$$f(x, y, z) = r^{-3}.$$

As $r \rightarrow \infty$ we have $r^{-3} \rightarrow 0^+$, so values can be arbitrarily close to 0. As $r \rightarrow 0^+$ (approaching the origin but not equal to it), we have $r^{-3} \rightarrow +\infty$. Hence

$$\text{ran}(f) = (0, \infty).$$

Example: Example 2: A Vector-Valued Function on $\mathbb{R}^6$:

Define $g : \mathbb{R}^6 \rightarrow \mathbb{R}^2$ by

$$g(x_1, x_2, x_3, x_4, x_5, x_6) = \left(x_1 x_2 x_3 x_4 x_5 x_6, \sqrt{x_1^2 + x_2^2} \right).$$

Step 1: Domain (Rigorous): The first component $x_1 x_2 x_3 x_4 x_5 x_6$ is defined for all real inputs. The second component is $\sqrt{x_1^2 + x_2^2}$. Since $x_1^2 + x_2^2 \geq 0$ for all reals, the square root is defined. Therefore,

$$\text{dom}(g) = \mathbb{R}^6.$$

Step 2: Codomain and Well-Definedness: For each $(x_1, \dots, x_6) \in \mathbb{R}^6$, the output is an ordered pair of real numbers, so $g(x) \in \mathbb{R}^2$ and $g : \mathbb{R}^6 \rightarrow \mathbb{R}^2$ is well-defined.

Step 3: Range (Partial But Informative Description): The second coordinate satisfies

$$\sqrt{x_1^2 + x_2^2} \in [0, \infty).$$

Indeed, it is exactly the Euclidean norm of $(x_1, x_2) \in \mathbb{R}^2$:

$$\sqrt{x_1^2 + x_2^2} = \|(x_1, x_2)\|.$$

The first coordinate can be any real number (positive, negative, or zero), since by choosing suitable inputs one can realize any sign and magnitude. Thus the range is contained in

$$\text{ran}(g) \subseteq \mathbb{R} \times [0, \infty),$$

and in fact many points of $\mathbb{R} \times [0, \infty)$ are attained.

1.6 A Short Lemma Connecting to Later Geometry

Proposition: Injective, Surjective, Bijective (Optional Foundational Terms)

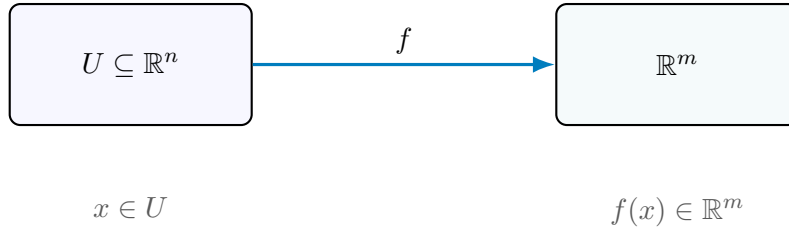
Let $f : U \rightarrow V$.

- f is *injective* if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.
- f is *surjective* if $\text{ran}(f) = V$.
- f is *bijective* if it is both injective and surjective.

Remark: Why This Matters Here

Later, when we study graphs and level sets, injectivity/surjectivity help explain whether different inputs can share the same output and whether every output level is actually achieved.

1.7 A Mapping Diagram



2 Graphs of Functions

2.1 Definition of the Graph

Definition: Graph in $\mathbb{R}^{n+1}$

Let $U \subseteq \mathbb{R}^n$ and $f : U \rightarrow \mathbb{R}$. The *graph* of f is the subset of $\mathbb{R}^{n+1} = \mathbb{R}^n \times \mathbb{R}$ defined by

$$\text{graph}(f) := \{(x, f(x)) \in U \times \mathbb{R} : x \in U\}.$$

In coordinates, if $x = (x_1, \dots, x_n)$, then

$$\text{graph}(f) = \{(x_1, \dots, x_n, f(x_1, \dots, x_n)) \in \mathbb{R}^{n+1} : (x_1, \dots, x_n) \in U\}.$$

Remark: Geometric Meaning and Dimension

- For $n = 1$, $\text{graph}(f) \subset \mathbb{R}^2$ is a plane curve.
- For $n = 2$, $\text{graph}(f) \subset \mathbb{R}^3$ is typically a surface.
- For $n \geq 3$, $\text{graph}(f) \subset \mathbb{R}^{n+1}$ is still well-defined as a set, even though direct visualization becomes difficult.

2.2 Basic Lemmas About Graphs

Proposition: Uniqueness of The Function Determined by its Graph

Let $f, g : U \rightarrow \mathbb{R}$. If $\text{graph}(f) = \text{graph}(g)$, then $f = g$.

Proof. Fix $x \in U$. Then $(x, f(x)) \in \text{graph}(f) = \text{graph}(g)$, hence $(x, f(x)) \in \text{graph}(g)$. By definition of $\text{graph}(g)$, the only point in $\text{graph}(g)$ with first coordinate x is $(x, g(x))$. Therefore $f(x) = g(x)$. Since x was arbitrary, $f = g$ on U . $\square$

Proposition: Fiber of a Graph is Always a Singleton

Let $f : U \rightarrow \mathbb{R}$ and set $G = \text{graph}(f)$. Then for every $x \in U$,

$$G_x = \{f(x)\}.$$

Proof. By definition,

$$G_x = \{y \in \mathbb{R} : (x, y) \in \text{graph}(f)\}.$$

But $(x, y) \in \text{graph}(f)$ holds if and only if $y = f(x)$. Hence $G_x = \{f(x)\}$. $\square$

2.3 Characterization Theorem: When is a Set a Graph?

Theorem: Graph Characterization Via Fibers (Vertical-Line Principle)

Let $U \subseteq \mathbb{R}^n$ and let $G \subseteq U \times \mathbb{R}$. Then $G = \text{graph}(f)$ for some function $f : U \rightarrow \mathbb{R}$ if and only if

$$\forall x \in U, \quad G_x \text{ is a singleton.}$$

Equivalently,

$$\forall x \in U \exists! y \in \mathbb{R} \text{ such that } (x, y) \in G.$$

Proof. ($\Rightarrow$) If $G = \text{graph}(f)$, then by the previous proposition $G_x = \{f(x)\}$ for all $x \in U$, hence G_x is a singleton.

($\Leftarrow$) Assume that for each $x \in U$, the fiber G_x is a singleton. Define $f : U \rightarrow \mathbb{R}$ by letting $f(x)$ be the unique real number in G_x . This definition is well-defined because G_x contains exactly one element. Now we prove $G = \text{graph}(f)$.

First, if $(x, y) \in G$, then $y \in G_x = \{f(x)\}$, so $y = f(x)$ and thus $(x, y) \in \text{graph}(f)$. Hence $G \subseteq \text{graph}(f)$.

Conversely, if $(x, f(x)) \in \text{graph}(f)$, then by definition of $f(x)$ we have $f(x) \in G_x$, which means $(x, f(x)) \in G$. Hence $\text{graph}(f) \subseteq G$.

Therefore $G = \text{graph}(f)$. □

2.4 Rigorous Solved Example (Graph Test)

Example: Example: $G = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}$ is not the graph of a function $\mathbb{R} \rightarrow \mathbb{R}$

Consider the set

$$G = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}.$$

Decide whether G is the graph of some function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Solution. To be the graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$, the set $G \subseteq \mathbb{R} \times \mathbb{R}$ must satisfy: for each $x \in \mathbb{R}$, there exists a *unique* $y \in \mathbb{R}$ such that $(x, y) \in G$.

Fix $x = -1$. Then $(x, y) \in G$ would require $y^2 = -1$, which has no real solution. Hence the fiber

$$G_{-1} = \{y \in \mathbb{R} : (-1, y) \in G\} = \emptyset$$

is empty, so the existence condition fails. Therefore, G cannot be the graph of any function $\mathbb{R} \rightarrow \mathbb{R}$.

Even if one restricts the domain to $U = [0, \infty)$, uniqueness fails: for each $x > 0$, the equation $y^2 = x$ has two distinct solutions $y = \sqrt{x}$ and $y = -\sqrt{x}$, so

$$G_x = \{\sqrt{x}, -\sqrt{x}\} \quad (x > 0),$$

which is not a singleton. Thus G is not the graph of a function $[0, \infty) \rightarrow \mathbb{R}$ either.

Remark: How to repair the set so it becomes a graph

If we restrict to the upper branch by requiring $y \geq 0$, then

$$G^+ := \{(x, y) \in \mathbb{R}^2 : y^2 = x, y \geq 0\}$$

is the graph of the function $f(x) = \sqrt{x}$ on $[0, \infty)$. Similarly, requiring $y \leq 0$ yields the graph of $f(x) = -\sqrt{x}$.

3 Level Sets, Curves, and Surfaces

3.1 Motivation

When a function has several variables, its graph may live in a higher-dimensional space and becomes hard to picture directly. A powerful alternative is to study the *geometry of constant values* of the function.

Definition: Level Set

Let $U \subseteq \mathbb{R}^n$ and let $f : U \rightarrow \mathbb{R}$ be a scalar-valued function. For a real number $c \in \mathbb{R}$, the **level set** of f at level c is

$$L_c(f) = \{x \in U : f(x) = c\}.$$

Remark: Terminology by Dimension

- If $n = 2$ and $f(x, y)$ is a function of two variables, then $L_c(f) \subseteq \mathbb{R}^2$ is often called a **level curve** or **contour line**.
- If $n = 3$ and $f(x, y, z)$ is a function of three variables, then $L_c(f) \subseteq \mathbb{R}^3$ is often called a **level surface**.
- A level set $L_c(f)$ always lies in the *domain space* $\mathbb{R}^n$ (not in $\mathbb{R}^{n+1}$). The graph of f is a subset of $\mathbb{R}^{n+1}$, but level sets are subsets of $\mathbb{R}^n$.

3.2 A Standard Geometric Picture

Take a function of three variables such as

$$f(x, y, z) = x^2 + y^2 + z^2.$$

Fix a constant $c \in \mathbb{R}$. The level set is

$$L_c(f) = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = c\}.$$

Classification.

- If $c < 0$, then $x^2 + y^2 + z^2 = c$ has no real solutions, so $L_c(f) = \emptyset$.
- If $c = 0$, then $x^2 + y^2 + z^2 = 0$ forces $x = y = z = 0$, so $L_0(f) = \{(0, 0, 0)\}$.
- If $c > 0$, then $x^2 + y^2 + z^2 = c$ is a sphere of radius $\sqrt{c}$ centered at the origin.

Proposition: Level surfaces of $x^2 + y^2 + z^2$

For $f(x, y, z) = x^2 + y^2 + z^2$, the level set $L_c(f)$ equals

$$L_c(f) = \begin{cases} \emptyset, & c < 0, \\ \{(0, 0, 0)\}, & c = 0, \\ \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = c\}, & c > 0, \end{cases}$$

and for $c > 0$ this is a sphere of radius $\sqrt{c}$.

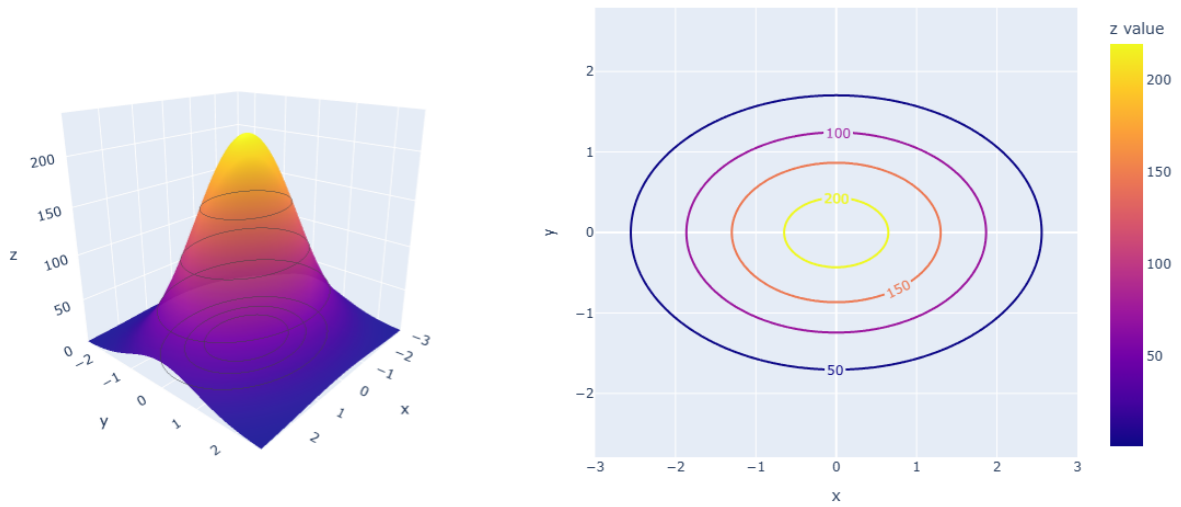


Figure 1: (a) Surface $z = h(x,y)$ with level curves (contours), (b) Contour map (top view): level curves $h(x,y)=c$

3.3 Contour Maps (Level Curves) for a Height Function

Often, a function of two variables

$$h : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto h(x, y)$$

is interpreted as a **height** above the xy -plane. Its graph is the surface $z = h(x, y)$. Instead of drawing the whole surface, one draws *contours*:

$$h(x, y) = c \iff (x, y) \in L_c(h).$$

Walking along a contour line keeps your height $h(x, y)$ constant. A family of contours gives a “topographic map” of the surface.

3.4 Example 1 (Rigorous): Constant Function and Its Level Curves

Example: Example 1: $f(x, y) = 2$

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = 2 \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

(1) Graph. By definition,

$$\text{graph}(f) = \{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}.$$

Since $f(x, y) = 2$ everywhere,

$$\text{graph}(f) = \{(x, y, z) \in \mathbb{R}^3 : z = 2\}.$$

Thus the graph is the **horizontal plane** $z = 2$ in $\mathbb{R}^3$.

(2) Level curve / level set at value $c \in \mathbb{R}$. By definition,

$$L_c(f) = \{(x, y) \in \mathbb{R}^2 : f(x, y) = c\}.$$

But $f(x, y)$ is identically 2, so the equation $f(x, y) = c$ becomes

$$2 = c.$$

Hence there are two cases:

Case 1: $c = 2$. Then $f(x, y) = 2$ holds for *every* $(x, y) \in \mathbb{R}^2$, so

$$L_2(f) = \mathbb{R}^2.$$

Case 2: $c \neq 2$. Then $2 = c$ is false, so there is *no* point (x, y) satisfying $f(x, y) = c$; thus

$$L_c(f) = \emptyset \quad (c \neq 2).$$

$$L_c(f) = \begin{cases} \mathbb{R}^2, & c = 2, \\ \emptyset, & c \neq 2. \end{cases}$$

3.5 Example 2 (Rigorous): Linear Function and Its Level Curves

Example: Example 2: $f(x, y) = x + y + 2$

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = x + y + 2.$$

(1) Graph. A point $(x, y, z) \in \mathbb{R}^3$ lies on $\text{graph}(f)$ if and only if

$$z = f(x, y) = x + y + 2.$$

Therefore,

$$\text{graph}(f) = \{(x, y, z) \in \mathbb{R}^3 : z = x + y + 2\},$$

which is a **plane** in $\mathbb{R}^3$.

(2) Intersection with the xy -plane. The xy -plane is given by $z = 0$. Intersecting with $z = x + y + 2$ gives

$$0 = x + y + 2 \iff y = -x - 2.$$

So the plane intersects the xy -plane along the **line** $y = -x - 2$.

(3) Level curve at level $c \in \mathbb{R}$. By definition,

$$L_c(f) = \{(x, y) \in \mathbb{R}^2 : f(x, y) = c\} = \{(x, y) \in \mathbb{R}^2 : x + y + 2 = c\}.$$

Solve algebraically for y :

$$x + y + 2 = c \iff y = c - 2 - x \iff y = -x + (c - 2).$$

Hence

$$L_c(f) = \{(x, y) \in \mathbb{R}^2 : y = -x + (c - 2)\}.$$

This is a straight line with slope -1 . Different values of c yield a family of parallel lines.

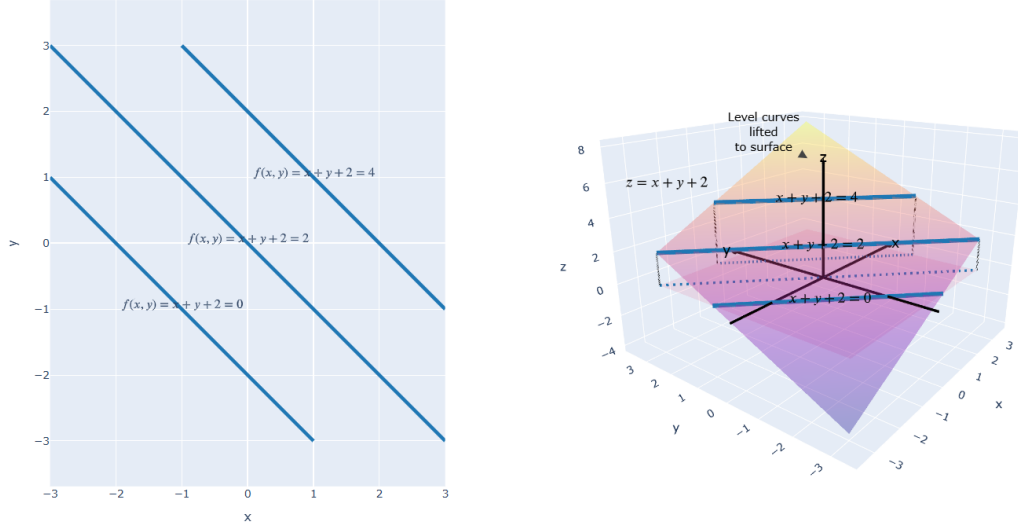


Figure 2: Level Curves of $f(x, y) = x + y + 2$ And Their Relation to The Graph $z = x + y + 2$

4 Level Curves and Surfaces

4.1 Definitions and Basic Principles

Definition: Level set, Level curve, Level surface

Let $U \subseteq \mathbb{R}^n$ and let $f : U \rightarrow \mathbb{R}$. For a constant $c \in \mathbb{R}$, the *level set of value c* is

$$L_c(f) := \{x \in U : f(x) = c\}.$$

- If $n = 2$, the set $L_c(f) \subseteq \mathbb{R}^2$ is called a *level curve* (or *contour*).
- If $n = 3$, the set $L_c(f) \subseteq \mathbb{R}^3$ is called a *level surface*.

Definition: Horizontal Slicing Principle

Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and consider its graph

$$\text{graph}(f) = \{(x, z) \in U \times \mathbb{R} : z = f(x)\} \subseteq \mathbb{R}^{n+1}.$$

Fix $c \in \mathbb{R}$ and let $H_c := \{(x, z) \in \mathbb{R}^{n+1} : z = c\}$ be the horizontal hyperplane. Then

$$\text{graph}(f) \cap H_c = \{(x, c) : x \in L_c(f)\}.$$

Proof. A point $(x, z) \in \text{graph}(f) \cap H_c$ satisfies both $z = f(x)$ and $z = c$, hence $f(x) = c$ and $z = c$. So $(x, z) = (x, c)$ with $x \in L_c(f)$. Conversely, if $x \in L_c(f)$, then $f(x) = c$ and so $(x, c) \in \text{graph}(f)$ and also $(x, c) \in H_c$. $\square$

Theorem: Contours Determine The Shape (Informal But Precise)

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the family $\{L_c(f)\}_{c \in \mathbb{R}}$ encodes the geometry of the surface $z = f(x, y)$: each level curve $L_c(f)$ is the projection to the xy -plane of the horizontal slice

$$\text{graph}(f) \cap \{z = c\}.$$

Thus a contour map is the collection of all such projected slices.

Theorem: Contours Determine the Shape (Formal Statement)

Let $U \subseteq \mathbb{R}^2$ and let $f : U \rightarrow \mathbb{R}$ be a function. Define the graph

$$\text{graph}(f) := \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in U, z = f(x, y)\},$$

and for each $c \in \mathbb{R}$ define the horizontal plane

$$H_c := \{(x, y, z) \in \mathbb{R}^3 : z = c\},$$

and the level set (level curve)

$$L_c(f) := \{(x, y) \in U : f(x, y) = c\}.$$

Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection map $\pi(x, y, z) = (x, y)$. Then for every $c \in \mathbb{R}$,

$$\pi(\text{graph}(f) \cap H_c) = L_c(f).$$

Equivalently, the horizontal slice of the surface at height c is

$$\text{graph}(f) \cap H_c = \{(x, y, c) : (x, y) \in L_c(f)\},$$

and projecting this slice to the xy -plane recovers exactly the level curve $L_c(f)$.

Proof. Fix $c \in \mathbb{R}$.

Step 1: Identify the Slice $\text{graph}(f) \cap H_c$. Let $(x, y, z) \in \text{graph}(f) \cap H_c$. Then $(x, y, z) \in \text{graph}(f)$ implies $(x, y) \in U$ and $z = f(x, y)$. Also, $(x, y, z) \in H_c$ implies $z = c$. Hence $f(x, y) = c$, so $(x, y) \in L_c(f)$, and therefore

$$(x, y, z) = (x, y, c) \text{ with } (x, y) \in L_c(f).$$

This proves the inclusion

$$\text{graph}(f) \cap H_c \subseteq \{(x, y, c) : (x, y) \in L_c(f)\}.$$

Conversely, if $(x, y) \in L_c(f)$, then $(x, y) \in U$ and $f(x, y) = c$ by definition. Thus $(x, y, c) \in \text{graph}(f)$ (since $c = f(x, y)$) and also $(x, y, c) \in H_c$ (since its z -coordinate is c). Therefore $(x, y, c) \in \text{graph}(f) \cap H_c$, proving the reverse inclusion and hence

$$\text{graph}(f) \cap H_c = \{(x, y, c) : (x, y) \in L_c(f)\}.$$

Step 2: Project the slice to $\mathbb{R}^2$. Apply $\pi(x, y, z) = (x, y)$ to both sides. Using the description from Step 1,

$$\pi(\text{graph}(f) \cap H_c) = \pi(\{(x, y, c) : (x, y) \in L_c(f)\}) = \{(x, y) : (x, y) \in L_c(f)\} = L_c(f).$$

Since c was arbitrary, the identity holds for all $c \in \mathbb{R}$. $\square$

4.2 Example: The Quadratic Function $f(x, y) = x^2 + y^2$

Example: Example (Quadratic Height Function)

Describe the graph of the function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = x^2 + y^2.$$

Also describe its level curves.

4.2.1 Solution

(1) **Graph of f .** The graph is

$$\text{graph}(f) = \{(x, y, z) \in \mathbb{R}^3 : z = x^2 + y^2\}.$$

Let $r := \sqrt{x^2 + y^2} \geq 0$. Then $z = r^2$ and hence $z \geq 0$ on the graph. Moreover, for each fixed $z \geq 0$, the cross-section of the graph by the plane $z = \text{constant}$ is a circle in the xy -plane of radius $\sqrt{z}$. Therefore the surface is a *paraboloid of revolution* opening upward with vertex at $(0, 0, 0)$.

(2) **Level curves.** Fix $c \in \mathbb{R}$. The level set is

$$L_c(f) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = c\}.$$

If $c < 0$, there is no real solution since $x^2 + y^2 \geq 0$, hence $L_c(f) = \emptyset$. If $c = 0$, then $x^2 + y^2 = 0$ implies $x = y = 0$, so $L_0(f) = \{(0, 0)\}$. If $c > 0$, then $x^2 + y^2 = c$ is a circle centered at the origin of radius $\sqrt{c}$:

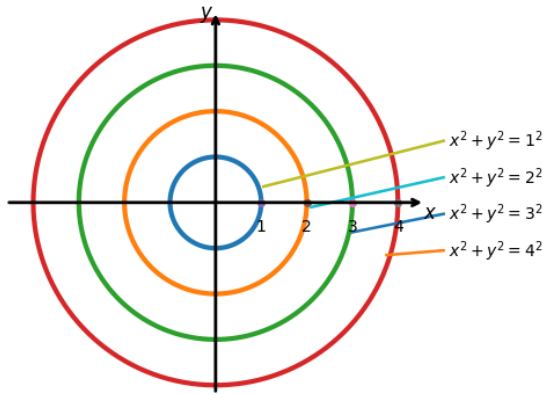
$$L_c(f) = \{(x, y) : \sqrt{x^2 + y^2} = \sqrt{c}\}.$$

Thus the contours are concentric circles, with radius increasing as c increases.

Remark: Interpretation

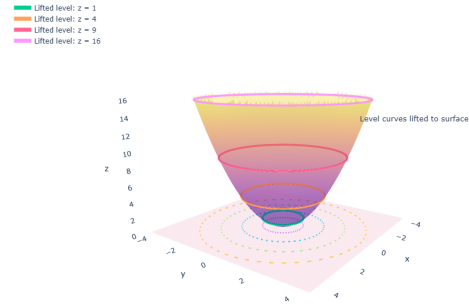
The contour map of $f(x, y) = x^2 + y^2$ consists of circles $x^2 + y^2 = 1, 4, 9, 16, \dots$. These lift to horizontal circles on the surface at heights $z = 1, 4, 9, 16, \dots$, forming the standard “stacked-ring” picture of the paraboloid.

Some level curves of $f(x, y) = x^2 + y^2$ (circles $x^2 + y^2 = c$)



(a) Level curves of $f(x, y) = x^2 + y^2$.

Level curves lifted to the surface: $z = x^2 + y^2$



(b) Lifted level curves on the surface $z = x^2 + y^2$.

Figure 3: (a) Contour map in the xy -plane. (b) The same level curves lifted to the graph $z = x^2 + y^2$ at heights $z = c$.

5 The Method of Sections

5.1 Foundational Definitions

Definition: Section of a Surface by a Plane

Let $S \subseteq \mathbb{R}^3$ and let $P \subseteq \mathbb{R}^3$ be a plane. The *section of S by P* is the intersection

$$S \cap P.$$

When $S = \text{graph}(f)$ is the graph of a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, a *vertical plane* (parallel to the z -axis) produces a planar curve whose shape helps us infer the geometry of S .

Definition: Vertical Planes Through Coordinate Directions

A plane is called *vertical* if it is parallel to the z -axis. Two important families are

$$P_{y_0} := \{(x, y, z) \in \mathbb{R}^3 : y = y_0\}, \quad Q_{x_0} := \{(x, y, z) \in \mathbb{R}^3 : x = x_0\},$$

where $x_0, y_0 \in \mathbb{R}$ are constants.

5.2 A Rigorous Lemma for Graph Sections

Lemma Sections of a Graph by Coordinate Vertical Planes

Let $U \subseteq \mathbb{R}^2$ and let $f : U \rightarrow \mathbb{R}$. Write

$$\text{graph}(f) = \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in U, z = f(x, y)\}.$$

Fix $y_0 \in \mathbb{R}$ and define the slice-domain

$$U_{y_0} := \{x \in \mathbb{R} : (x, y_0) \in U\}.$$

Then

$$\text{graph}(f) \cap P_{y_0} = \{(x, y_0, f(x, y_0)) : x \in U_{y_0}\}.$$

Similarly, for fixed $x_0 \in \mathbb{R}$ with

$$U^{x_0} := \{y \in \mathbb{R} : (x_0, y) \in U\},$$

we have

$$\text{graph}(f) \cap Q_{x_0} = \{(x_0, y, f(x_0, y)) : y \in U^{x_0}\}.$$

Proof. We prove the first identity; the second is analogous.

($\subseteq$) Let $(x, y, z) \in \text{graph}(f) \cap P_{y_0}$. Then $(x, y, z) \in P_{y_0}$ implies $y = y_0$. Also $(x, y, z) \in \text{graph}(f)$ implies $(x, y) \in U$ and $z = f(x, y)$. Hence $(x, y_0) \in U$ so $x \in U_{y_0}$ and $z = f(x, y_0)$. Therefore $(x, y, z) = (x, y_0, f(x, y_0))$ with $x \in U_{y_0}$.

($\supseteq$) Let $x \in U_{y_0}$. Then $(x, y_0) \in U$, hence $(x, y_0, f(x, y_0)) \in \text{graph}(f)$ by definition. Also its y -coordinate equals y_0 , so it belongs to P_{y_0} . Thus it belongs to $\text{graph}(f) \cap P_{y_0}$. $\square$

Theorem: Method of Sections (Geometric Reconstruction Principle)

Let $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ and let $S = \text{graph}(f)$. For each constant y_0 , the section $S \cap P_{y_0}$ is the graph (in the plane $y = y_0$) of the one-variable function

$$g_{y_0} : U_{y_0} \rightarrow \mathbb{R}, \quad g_{y_0}(x) = f(x, y_0).$$

For each constant x_0 , the section $S \cap Q_{x_0}$ is the graph (in the plane $x = x_0$) of the one-variable function

$$h_{x_0} : U^{x_0} \rightarrow \mathbb{R}, \quad h_{x_0}(y) = f(x_0, y).$$

Therefore, by computing and sketching sufficiently many such planar sections, one can infer the global shape of S .

Proof. By the lemma,

$$S \cap P_{y_0} = \{(x, y_0, f(x, y_0)) : x \in U_{y_0}\} = \{(x, y_0, g_{y_0}(x)) : x \in U_{y_0}\},$$

which is precisely the graph of g_{y_0} inside the plane $y = y_0$. The $x = x_0$ case is identical. $\square$

5.3 Example 4: The Hyperbolic Paraboloid (saddle)

Example: Example 4

Describe the graph of the quadratic function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = x^2 - y^2.$$

Solution

(1) Identify key sections. Consider the vertical planes $y = 0$ and $x = 0$.

- In the plane $y = 0$:

$$z = f(x, 0) = x^2,$$

so the section is the parabola $z = x^2$ opening upward.

- In the plane $x = 0$:

$$z = f(0, y) = -y^2,$$

so the section is the parabola $z = -y^2$ opening downward.

These opposite curvatures along perpendicular directions imply the surface is a *saddle*.

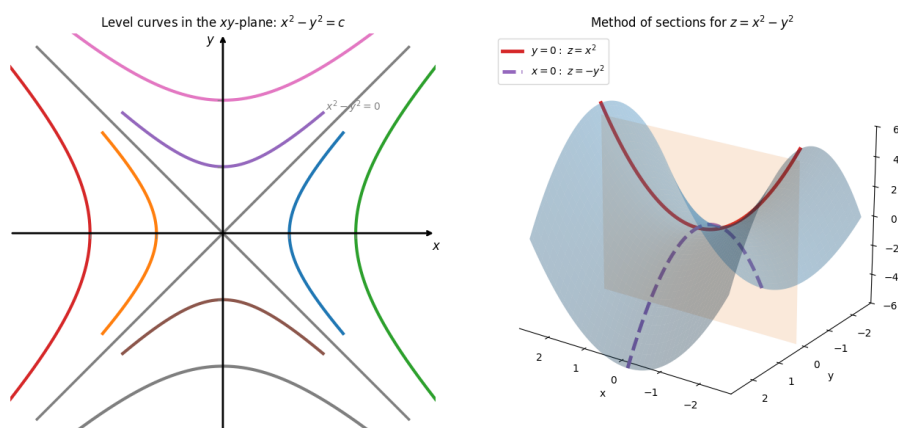
(2) Level Curves (contours). For $c \in \mathbb{R}$, the level curve in the xy -plane is

$$L_c = \{(x, y) \in \mathbb{R}^2 : x^2 - y^2 = c\}.$$

- If $c > 0$, then $x^2 - y^2 = c$ is a hyperbola opening along the x -axis.
- If $c = 0$, then $x^2 = y^2$, i.e. $y = \pm x$ (two intersecting lines).
- If $c < 0$, then $y^2 - x^2 = -c$ is a hyperbola opening along the y -axis.

Remark: Why This is a Saddle (Precise Geometric criterion)

Along the line $y = 0$, the restriction is $x \mapsto x^2$ (convex, minimum at 0). Along the line $x = 0$, the restriction is $y \mapsto -y^2$ (concave, maximum at 0). Hence the origin is neither a local maximum nor a local minimum of f .



5.4 Example 5: Level Surfaces of $x^2 + y^2 + z^2$

Example: Example 5

Describe the level sets of

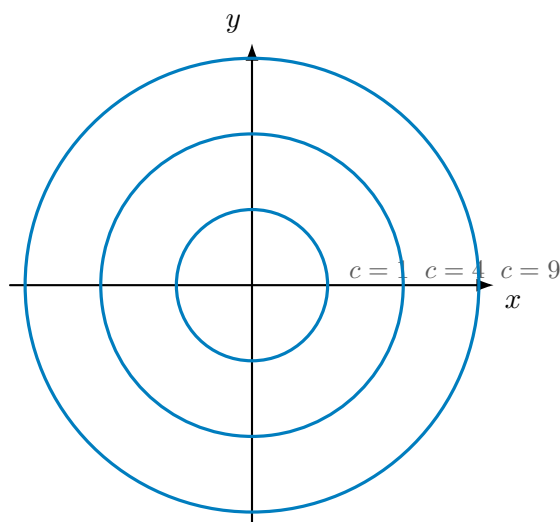
$$F : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad F(x, y, z) = x^2 + y^2 + z^2.$$

5.4.1 Solution

Fix $c \in \mathbb{R}$. The level set is

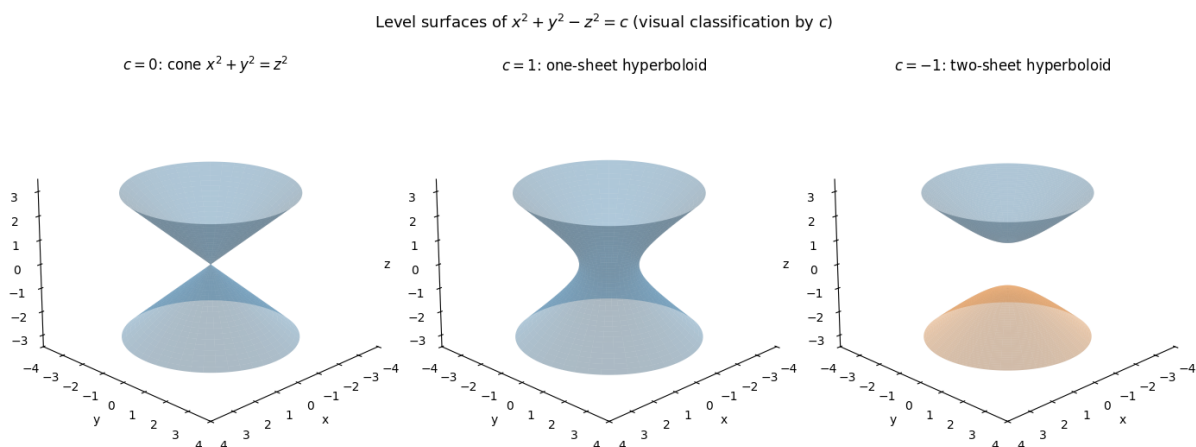
$$L_c(F) = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = c\}.$$

Since $x^2 + y^2 + z^2 \geq 0$, if $c < 0$ then $L_c(F) = \emptyset$. If $c = 0$, then $x^2 + y^2 + z^2 = 0$ implies $(x, y, z) = (0, 0, 0)$. If $c > 0$, then $x^2 + y^2 + z^2 = c$ is a sphere of radius $\sqrt{c}$ centered at the origin.



Spheres $x^2 + y^2 + z^2 = c$ project to circles of radius $\sqrt{c}$ in the xy -plane.

Figure 4: Schematic representation of level surfaces $x^2 + y^2 + z^2 = c$ (spheres).



5.5 Example 6: Level Surfaces of $x^2 + y^2 - z^2$

Example: Example 6

Describe the level surfaces of

$$G : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad G(x, y, z) = x^2 + y^2 - z^2.$$

5.5.1 Solution

Fix $c \in \mathbb{R}$. The level surface is

$$L_c(G) = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = c\}.$$

Write $r^2 := x^2 + y^2$ (cylindrical radius). Then the equation becomes

$$r^2 - z^2 = c.$$

- If $c = 0$, then $r^2 = z^2$, i.e. $r = |z|$: a double cone (opening along the z -axis).
- If $c > 0$, then $r^2 = z^2 + c \geq c$, so $r \geq \sqrt{c}$ for all z ; this is a *one-sheet hyperboloid* of revolution about the z -axis.
- If $c < 0$, write $c = -a^2$ with $a > 0$. Then $r^2 - z^2 = -a^2$ implies $z^2 = r^2 + a^2 \geq a^2$, so $|z| \geq a$; this yields two disconnected sheets: a *two-sheet hyperboloid*.

6 Method of Sections for Graphs of Functions of Three Variables

6.1 Graphs in $\mathbb{R}^4$ and why we use sections

Definition: Graph of a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$

Let $U \subseteq \mathbb{R}^3$ and let $f : U \rightarrow \mathbb{R}$. The *graph* of f is the subset of $\mathbb{R}^4$ defined by

$$\text{graph}(f) := \{(x, y, z, t) \in \mathbb{R}^4 : (x, y, z) \in U, t = f(x, y, z)\}.$$

Because $\mathbb{R}^4$ cannot be directly visualized, we study $\text{graph}(f)$ via *3-dimensional slices* obtained by intersecting $\text{graph}(f)$ with hyperplanes such as $\{y = y_0\}$ or $\{x = x_0\}$. Each such slice is itself a surface in a *three-dimensional* space and can therefore be drawn.

Definition: Slice (section) by a coordinate hyperplane

Fix $y_0 \in \mathbb{R}$. Define the hyperplane in $\mathbb{R}^4$

$$H_{y_0} := \{(x, y, z, t) \in \mathbb{R}^4 : y = y_0\}.$$

The $y = y_0$ section of $\text{graph}(f)$ is $\text{graph}(f) \cap H_{y_0}$.

6.2 A formal lemma describing coordinate sections**Lemma Coordinate section of a graph is the graph of a sliced function**

Let $U \subseteq \mathbb{R}^3$ and $f : U \rightarrow \mathbb{R}$. Fix $y_0 \in \mathbb{R}$ and define the slice-domain

$$U_{y_0} := \{(x, z) \in \mathbb{R}^2 : (x, y_0, z) \in U\}.$$

Define the *sliced function*

$$g_{y_0} : U_{y_0} \rightarrow \mathbb{R}, \quad g_{y_0}(x, z) := f(x, y_0, z).$$

Then

$$\text{graph}(f) \cap H_{y_0} = \{(x, y_0, z, g_{y_0}(x, z)) : (x, z) \in U_{y_0}\}.$$

Equivalently, after identifying $H_{y_0} \cong \mathbb{R}^3$ via $(x, y_0, z, t) \leftrightarrow (x, z, t)$, the set $\text{graph}(f) \cap H_{y_0}$ is precisely the 3D graph

$$t = g_{y_0}(x, z) \quad \text{in the coordinates } (x, z, t).$$

Proof. Fix $y_0 \in \mathbb{R}$.

($\subseteq$) Let $(x, y, z, t) \in \text{graph}(f) \cap H_{y_0}$. Then $(x, y, z, t) \in H_{y_0}$ implies $y = y_0$. Also $(x, y, z, t) \in \text{graph}(f)$ implies $(x, y, z) \in U$ and $t = f(x, y, z)$. Therefore $(x, y_0, z) \in U$, hence $(x, z) \in U_{y_0}$ by definition. Moreover,

$$t = f(x, y_0, z) = g_{y_0}(x, z).$$

Thus

$$(x, y, z, t) = (x, y_0, z, g_{y_0}(x, z)) \quad \text{with } (x, z) \in U_{y_0}.$$

($\supseteq$) Conversely, take any $(x, z) \in U_{y_0}$. Then $(x, y_0, z) \in U$ by definition of U_{y_0} . Hence

$$(x, y_0, z, g_{y_0}(x, z)) = (x, y_0, z, f(x, y_0, z)) \in \text{graph}(f).$$

Also its y -coordinate equals y_0 , so it lies in H_{y_0} . Therefore it lies in $\text{graph}(f) \cap H_{y_0}$.

Combining the two inclusions gives the desired equality. $\square$

Theorem: Method of Sections (precise reconstruction principle)

Let $f : U \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$. For each fixed value y_0 for which U_{y_0} is nonempty, the section $\text{graph}(f) \cap \{y = y_0\}$ is the 3D graph of the two-variable function $g_{y_0}(x, z) = f(x, y_0, z)$. Thus the family of all such sections (for varying y_0) provides systematic geometric information about $\text{graph}(f)$.

Proof. This is an immediate consequence of the lemma: each section is exactly a 3D graph in (x, z, t) -space. Varying y_0 varies the section, thereby revealing how the full 4D object $\text{graph}(f)$ changes with y . $\square$

6.3 Worked example in full detail: $f(x, y, z) = x^2 + y^2 - z^2$

Example: Example: a graph in $\mathbb{R}^4$ and the section $y = 0$

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be

$$f(x, y, z) = x^2 + y^2 - z^2.$$

1. Describe the section of $\text{graph}(f)$ by the hyperplane $y = 0$.
2. On this section, describe the curves obtained by intersecting with horizontal planes $t = c$.
3. Explain rigorously why the section surface is saddle-shaped.

Solution

(1) Compute the $y = 0$ section. Here $U = \mathbb{R}^3$, so $U_0 = \mathbb{R}^2$ (all (x, z) are allowed). Define the sliced function

$$g_0(x, z) = f(x, 0, z) = x^2 + 0^2 - z^2 = x^2 - z^2.$$

By the lemma,

$$\text{graph}(f) \cap \{y = 0\} = \{(x, 0, z, g_0(x, z)) : (x, z) \in \mathbb{R}^2\}.$$

Under the identification $(x, 0, z, t) \leftrightarrow (x, z, t)$, this is the 3D surface

$$t = x^2 - z^2 \quad \text{in } (x, z, t)\text{-space.}$$

This surface is a *hyperbolic paraboloid*.

(2) Horizontal slicing on the section gives level curves. Fix $c \in \mathbb{R}$. The intersection of the section surface with the horizontal plane $\{t = c\}$ in (x, z, t) -space is:

$$\{(x, z, t) : t = x^2 - z^2\} \cap \{(x, z, t) : t = c\}.$$

A point lies in the intersection if and only if $t = c$ and $t = x^2 - z^2$, hence

$$x^2 - z^2 = c.$$

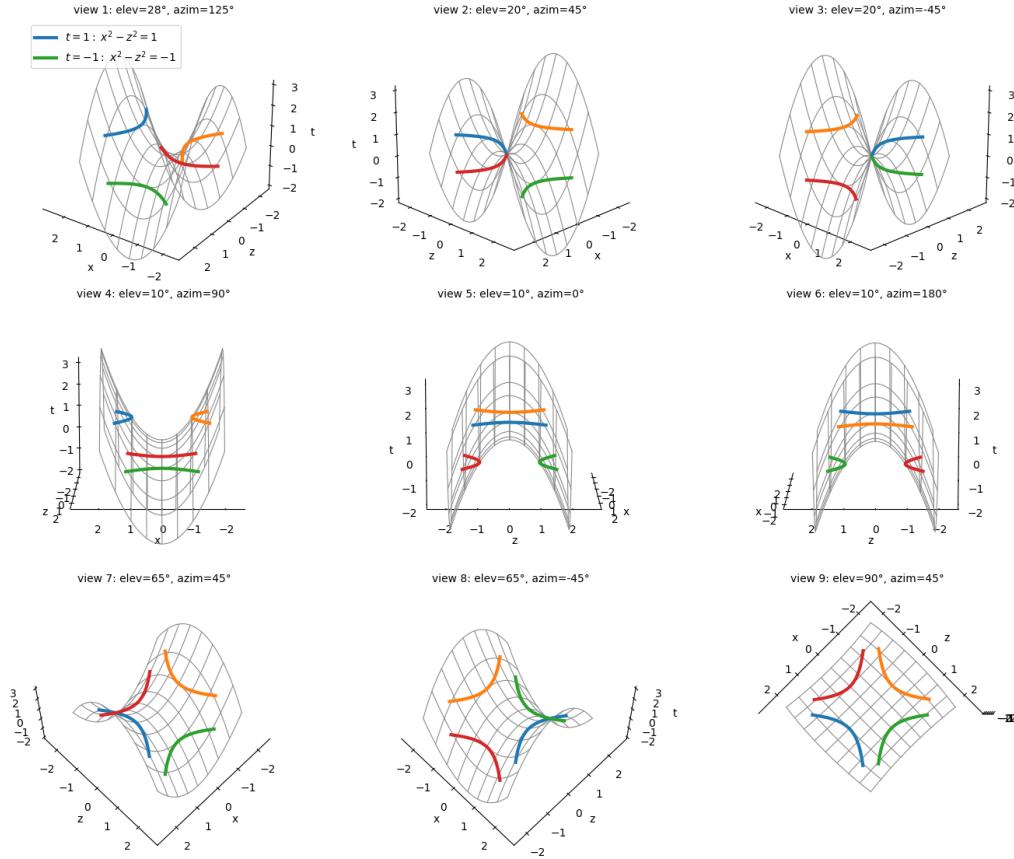
Therefore the slice curves are

$$C_c := \{(x, z) \in \mathbb{R}^2 : x^2 - z^2 = c\} \quad (\text{and lifted to height } t = c).$$

We classify these curves:

- If $c > 0$, then $\frac{x^2}{c} - \frac{z^2}{c} = 1$, a hyperbola opening along the x -direction.
- If $c = 0$, then $x^2 = z^2$, i.e. $z = \pm x$, a pair of lines through the origin.
- If $c < 0$, write $c = -a^2$ with $a > 0$. Then $z^2 - x^2 = a^2$, a hyperbola opening along the z -direction.

9 projections of the section surface $t = x^2 - z^2$ with slice curves $t = \pm 1$



(3) Saddle property via opposite curvature directions (rigorous). Consider restrictions of $g_0(x, z) = x^2 - z^2$ along the coordinate lines:

$$g_0(x, 0) = x^2 \quad \text{and} \quad g_0(0, z) = -z^2.$$

Thus, along the $z = 0$ line in the domain, the surface cross-section is the upward-opening parabola $t = x^2$, while along the $x = 0$ line it is the downward-opening parabola $t = -z^2$. Hence the origin is neither a local maximum nor a local minimum: it is a *saddle point*. In fact, for any $\varepsilon > 0$,

$$g_0(\varepsilon, 0) = \varepsilon^2 > 0 \quad \text{but} \quad g_0(0, \varepsilon) = -\varepsilon^2 < 0,$$

so in every neighborhood of $(0, 0)$ the function takes values both above and below 0.

Remark: Geometric meaning

The full graph $\text{graph}(f) \subset \mathbb{R}^4$ is encoded by the family of 3D surfaces

$$t = f(x, y, z) = x^2 + y^2 - z^2 \quad \text{restricted to fixed } y = y_0,$$

which yields

$$t = x^2 + y_0^2 - z^2.$$

Each y_0 shifts the saddle vertically by y_0^2 in the t -direction.

6.4 Practice Problems

Example: Practice Set A: Level Curves and Contour Geometry in $\mathbb{R}^2$

1. Let $f(x, y) = x^2 + y^2$.

- (a) Compute $L_c(f)$ for $c < 0$, $c = 0$, and $c > 0$.
- (b) Prove your classification using only algebra and the definition

$$L_c(f) = \{(x, y) \in \mathbb{R}^2 : f(x, y) = c\}.$$

- (c) For $c > 0$, compute the radius of $L_c(f)$ and show that $L_c(f)$ is connected.

2. Let $g(x, y) = x^2 - y^2$.

- (a) Describe $L_c(g)$ for $c > 0$, $c = 0$, and $c < 0$.
- (b) Prove that $L_0(g)$ is exactly the union of two lines.
- (c) Prove that for $c \neq 0$, $L_c(g)$ is a hyperbola and determine whether it opens in the x -direction or y -direction.

3. Let $h(x, y) = \sqrt{x^2 + y^2}$.

- (a) Compute $L_c(h)$ for $c < 0$, $c = 0$, and $c > 0$.
- (b) Compare $L_c(h)$ with $L_c(f)$ where $f(x, y) = x^2 + y^2$.
- (c) Prove that for $c \geq 0$,

$$L_c(h) = L_{c^2}(f).$$

4. (Non-circular contours) Let $p(x, y) = x^2 + 4y^2$.

- (a) Compute $L_c(p)$ for $c > 0$ and show it is an ellipse.
- (b) Find the intercepts on the axes and the semi-axis lengths.
- (c) Explain how changing the coefficient of y^2 affects the contour map geometrically.

5. (Shifted centers) Let $q(x, y) = (x - a)^2 + (y - b)^2$ where $(a, b) \in \mathbb{R}^2$ is fixed.

- (a) Compute $L_c(q)$ for $c > 0$ and identify its center and radius.
- (b) Prove that $L_0(q) = \{(a, b)\}$ and $L_c(q) = \emptyset$ for $c < 0$.

6. (A contour-map reasoning problem) Consider a function $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ whose level curves are the family of straight lines

$$x + y = c, \quad c \in \mathbb{R}.$$

Prove that there exists a one-variable function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$u(x, y) = \varphi(x + y).$$

(You may assume u is continuous if needed; state clearly where you use it.)

Example: Practice Set B: Graphs, Horizontal Slices, and the Contour–Surface Correspondence

1. Let $f(x, y) = x^2 + y^2$ and consider its graph $z = f(x, y)$ in $\mathbb{R}^3$.
 - (a) Show that the horizontal slice $\text{graph}(f) \cap \{z = c\}$ is empty for $c < 0$, is a point for $c = 0$, and is a circle for $c > 0$.
 - (b) For $c > 0$, compute the radius of the circle in $\mathbb{R}^3$.
 - (c) Prove that the projection of $\text{graph}(f) \cap \{z = c\}$ onto the xy -plane equals the level curve $L_c(f)$.
2. Let $g(x, y) = x^2 - y^2$ and consider its graph $z = g(x, y)$.
 - (a) Show that $\text{graph}(g) \cap \{z = c\}$ projects to the level curve $x^2 - y^2 = c$ in the xy -plane.
 - (b) Use this to explain (rigorously) why the contour map of g determines how the surface bends.
3. (Formal statement to prove) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and define the projection $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $\pi(x, y, z) = (x, y)$. Prove that for every $c \in \mathbb{R}$,

$$L_c(f) = \pi(\text{graph}(f) \cap \{z = c\}).$$

Your proof must be set-theoretic: prove both inclusions.

4. (Vertical line test, set version) Let $U \subseteq \mathbb{R}$ and $S \subseteq U \times \mathbb{R}$. Prove that S is the graph of some function $f : U \rightarrow \mathbb{R}$ if and only if: for each $x \in U$ there exists a *unique* $y \in \mathbb{R}$ such that $(x, y) \in S$.
5. (Projection ambiguity) Give an example of two different subsets $A, B \subset \mathbb{R}^3$ such that

$$\pi(A) = \pi(B) \subset \mathbb{R}^2$$

but $A \neq B$. Explain in one paragraph why this shows that a single projection can lose 3D information.

Example: Practice Set C: Method of Sections and Slices for Functions of Two Variables

1. For $f(x, y) = x^2 - y^2$, compute and sketch the sections of the graph $z = f(x, y)$ in the planes $y = 1$ and $x = 1$. Write each section as an explicit parabola in the appropriate plane.
2. For $f(x, y) = x^2 - y^2$, classify the level curves $x^2 - y^2 = c$ for

$$c \in \{-4, -1, 0, 1, 4\},$$

and prove each classification. Your proof should explicitly identify whether each curve is a hyperbola, a pair of lines, or empty.

3. (Saddle test by sections) Let $f(x, y) = x^2 - y^2$. Prove that the origin is a saddle point by restricting f to the lines $y = 0$ and $x = 0$, and interpreting these restrictions as planar sections of the graph.
4. (A section that becomes a line) Let $f(x, y) = x + y + 2$. Show that for each fixed $y = y_0$ the section of the graph $z = x + y + 2$ by the plane $y = y_0$ is a straight line in (x, z) -coordinates. Similarly, show that the horizontal slice at $z = c$ projects to the line $x + y = c - 2$ in the xy -plane.
5. (Design your own contour map) Construct a function $f(x, y)$ such that all level curves are circles but the spacing between consecutive level curves (for $c, c + 1$) increases as you move away from the origin. Explain your choice.

Example: Practice Set D: Level Surfaces in $\mathbb{R}^3$ (Spheres, Cones, Hyperboloids)

1. Let $F(x, y, z) = x^2 + y^2 + z^2$.
 - (a) Prove that for $c > 0$, $L_c(F)$ is a sphere of radius $\sqrt{c}$ centered at the origin.
 - (b) Prove that $L_0(F) = \{(0, 0, 0)\}$ and that $L_c(F) = \emptyset$ for $c < 0$.
 - (c) Give a proof using the Euclidean norm $\|(x, y, z)\| = \sqrt{x^2 + y^2 + z^2}$.
2. Let $G(x, y, z) = x^2 + y^2 - z^2$.
 - (a) Classify $L_c(G)$ for $c > 0$, $c = 0$, $c < 0$ (name each surface).
 - (b) Prove rigorously that if $c < 0$, then $L_c(G)$ has *two connected components*.
 - (c) Hint for (b): show that on $L_c(G)$ with $c < 0$, one must have $|z| \geq \sqrt{-c}$ and use the sign of z to separate components.
3. (Cross-sections in cylindrical coordinates) Let $H(x, y, z) = x^2 + y^2 - z^2$ and define $r = \sqrt{x^2 + y^2}$. Show that the equation $H = c$ becomes

$$r^2 - z^2 = c.$$

Sketch in the (r, z) -plane the cases $c = 0$, $c > 0$, and $c < 0$, and explain how revolving these curves about the z -axis produces the corresponding 3D surfaces.

4. (Sections of a 4D graph) Let $f(x, y, z) = x^2 + y^2 - z^2$ and consider $\text{graph}(f) \subset \mathbb{R}^4$ with coordinates (x, y, z, t) . Fix $y = y_0$ and prove that the resulting section is the surface

$$t = x^2 + y_0^2 - z^2 \quad \text{in } (x, z, t)\text{-space.}$$

Explain how varying y_0 changes the section geometrically.

5. (Challenge: connectedness) For $G(x, y, z) = x^2 + y^2 - z^2$, prove that $L_c(G)$ is connected when $c \geq 0$. (You may use a path construction between two points on the surface.)

Example: Practice Set E: Computation & Visualization Tasks (Mathematics + Python)

1. Using Python (Matplotlib), reproduce a contour map for $g(x, y) = x^2 - y^2$ for level values

$$c \in \{-4, -1, 0, 1, 4\}.$$

In your write-up, explain how the sign of c determines the orientation of the hyperbolas.

2. Using Python (Plotly), plot the surface $z = x^2 - y^2$ and overlay the lifted level curves for $c \in \{-4, -1, 0, 1, 4\}$. Explain precisely why the lifted curves lie on the surface.
3. Reproduce the 9-view projection grid for the saddle section $t = x^2 - z^2$ (as we did), but now highlight $t = 0$ as well. Explain how the curve $t = 0$ corresponds to the two ruling lines $z = \pm x$ on the surface.
4. Choose a new function (your choice) among:

$$x^2 + 4y^2, \quad (x - 1)^2 + (y + 2)^2, \quad \sin x + \cos y, \quad e^{-(x^2 + y^2)}.$$

Produce (i) a contour plot, (ii) a 3D surface plot, and (iii) a short paragraph explaining what the contour spacing tells you about steepness.

7 Conclusion

In this lecture note we developed a unified geometric understanding of real-valued functions by linking algebraic definitions, set-theoretic constructions, and visualization. A function $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ becomes a geometric object through its graph, and in the scalar-valued case ($m = 1$) the graph $\text{graph}(f) = \{(x, f(x)) : x \in U\} \subset \mathbb{R}^{n+1}$ provides a precise bridge from analysis to geometry. Since higher-dimensional graphs (for instance, $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ with $\text{graph}(f) \subset \mathbb{R}^4$) are not directly drawable, we relied on the method of sections: by fixing one coordinate (e.g. $y = y_0$) we obtained a three-dimensional section that is itself the graph of a sliced function, thus allowing rigorous visualization in a lower-dimensional space. Level sets $L_c(f) = \{x \in U : f(x) = c\}$ were treated as the structural “skeleton” of a graph: in two variables they appear as level curves (contours) and in three variables as level surfaces, and they arise naturally as projections of horizontal slices of the graph. The worked example $f(x, y, z) = x^2 + y^2 - z^2$ illustrated the full workflow: the section $y = 0$ yields the saddle surface $t = x^2 - z^2$ in (x, z, t) -space, while the intersections with $t = c$ produce the hyperbolas $x^2 - z^2 = c$, making the geometry of the surface transparent. Finally, we emphasized that every rendered 3D image is a projection depending on the camera view; therefore, multiple viewpoints (oblique, side, and top-down) are essential for correctly reading height, curvature, and slicing geometry. Altogether, the theory and visuals form a single coherent pipeline: *function* $\Rightarrow$ *graph* $\Rightarrow$ *slices/level sets* $\Rightarrow$ *rigorous geometric interpretation*.